

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but **do not open this question paper until you are ready to begin.**

This paper contains **27 multiple-choice questions**. There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all of the questions. Each question is worth one mark.

For each question, choose the one option you consider correct. If you make a mistake, erase thoroughly and try again.

You must complete the paper within the time limit.

You can use the question paper for rough working. Dictionaries and calculators are **NOT permitted**.

Please wait until you are ready before turning this page.

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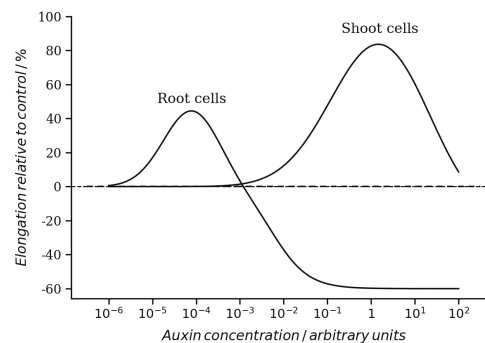
Biology

- 1 The graph shows the effect of auxin concentration on the elongation of root cells and shoot cells, expressed as the percentage change relative to an untreated control.

A germinating seedling is placed horizontally in the dark. Gravity causes auxin to accumulate on the lower side of both the root and the shoot, resulting in an auxin concentration of 10^{-1} arbitrary units on the lower side and 10^{-4} arbitrary units on the upper side in both organs.

Which of the following statements is/are correct?

- 1 The lower side of the shoot elongates faster than its upper side.
- 2 The lower side of the root elongates faster than its upper side.
- 3 The shoot will bend upwards and the root will bend downwards.



- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 2 A bacterium and a typical animal cell are both undergoing asexual cell division. The bacterium contains a single circular DNA molecule, while the animal cell contains multiple linear chromosomes.

Which row in the table correctly describes the mechanism of genetic material separation in each cell?

	Bacterium	Animal Cell
A	Spindle fibres pull DNA to opposite poles	Spindle fibres pull DNA to opposite poles
B	Replicated DNA separates as the cell elongates	Spindle fibres pull sister chromatids apart
C	Replicated DNA separates as the cell elongates	Spindle fibres pull homologous chromosomes apart
D	Sister chromatids are pulled apart by spindle fibres	Spindle fibres pull homologous chromosomes apart
E	Homologous chromosomes are pulled apart by spindle fibres	Spindle fibres pull sister chromatids apart

- 3 During pregnancy in a mammal, exchange of substances occurs between maternal blood and fetal blood across the placenta.

Which row in the table correctly identifies the substances that are present at a higher concentration in the blood of the umbilical vein compared to the blood of the umbilical artery?

Key:

✓ = higher concentration in the umbilical vein

X = not a higher concentration in the umbilical vein

	glucose	carbon dioxide	urea
A	✓	✓	✓
B	✓	✓	X
C	✓	X	✓
D	✓	X	X
E	X	✓	✓
F	X	✓	X
G	X	X	✓
H	X	X	X

- 4 An investigation was carried out to study the rate of protein synthesis in two different strains of a yeast species, Strain X and Strain Y.

Within each strain, all yeast cells were genetically identical clones.

Samples containing equal numbers of cells from each strain were grown in nutrient solutions with different concentrations of glucose (G_1 , G_2 , and G_3 , where $G_1 < G_2 < G_3$). All other environmental conditions (temperature, pH, dissolved oxygen, and amino acid concentrations) were kept constant and identical for all cultures.

After a set period of time, the mean rate of protein synthesis per cell was measured for each culture. The results are shown in the graph.

Which of the following statements is/are correct?

- 1 The difference in the mean rate of protein synthesis in Strain X between G_1 and G_2 is due to environmental variation.
- 2 The difference in the mean rate of protein synthesis between Strain X and Strain Y at G_2 is due to genetic variation.
- 3 The failure of the mean rate of protein synthesis to increase in Strain X between G_2 and G_3 could be caused by an environmental factor other than glucose becoming limiting.



- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

5 Which of the statements about somatic gene therapy used to treat a human genetic disorder caused by a faulty recessive allele are correct?

- 1 A vector is used to introduce a functional copy of the allele into target body cells.
- 2 The introduced allele is incorporated into the genomes of both somatic cells and gametes.
- 3 The functional protein is synthesised using the ribosomes of the patient's own cells.
- 4 Successful treatment ensures that the patient cannot pass the faulty allele to any biological offspring.

- A** 1 and 2 only
B 1 and 3 only
C 2 and 4 only
D 3 and 4 only
E 1, 2 and 3 only
F 1, 3 and 4 only
G 2, 3 and 4 only
H 1, 2, 3 and 4

6 Three different mechanisms of transport across a cell membrane, X, Y and Z, are compared in the table.

Mechanism of membrane transport	Transports substances across a cell membrane	Requires transmembrane protein carriers or channels	Can result in net movement against a concentration gradient	Requires energy directly from ATP hydrolysis
X	yes	yes	no	no
Y	yes	no	no	no
Z	yes	yes	yes	yes

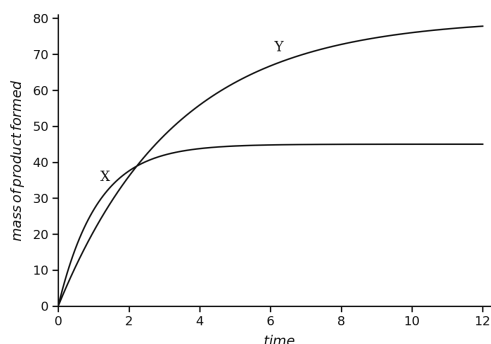
Which row in the table correctly identifies mechanisms X, Y and Z?

	X	Y	Z
A	facilitated diffusion	simple diffusion	active transport
B	facilitated diffusion	active transport	simple diffusion
C	simple diffusion	facilitated diffusion	active transport
D	simple diffusion	active transport	facilitated diffusion
E	active transport	facilitated diffusion	simple diffusion
F	active transport	simple diffusion	facilitated diffusion

- 7 The graph shows the total mass of product formed over time during an enzyme-catalysed reaction carried out under two different conditions, X and Y.

Which of the following statements could explain the difference between curve X and curve Y?

- 1 Condition X had a higher enzyme concentration and a lower initial mass of substrate than Condition Y.
- 2 Condition X was at a higher temperature that caused progressive thermal denaturation of the enzyme over time, whereas Condition Y was at a lower temperature at which the enzyme remained stable.
- 3 Condition X contained a competitive inhibitor, whereas Condition Y contained no inhibitor.
- 4 Condition X was at a pH further from the enzyme's optimum pH than Condition Y.



- A 1 only
- B 2 only
- C 1 and 2 only
- D 1 and 3 only
- E 2 and 4 only
- F 3 and 4 only
- G 1, 2 and 3 only
- H 1, 2, 3 and 4

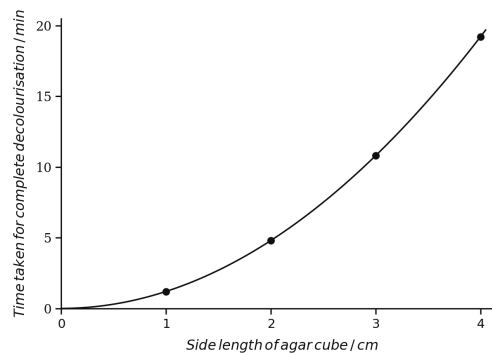
- 8 An investigation was carried out to model the effect of cell size on the rate of exchange of substances by diffusion.

Agar cubes of different side lengths were prepared containing dilute sodium hydroxide and the indicator phenolphthalein (which is pink in alkaline conditions). The cubes were placed into separate beakers containing equal volumes of dilute hydrochloric acid at the same concentration and temperature. The acid diffused into the cubes, neutralising the alkali and causing the cubes to decolourise.

The time taken for each cube to completely decolourise was recorded. The graph shows the results of this experiment.

Which of the following statements about the investigation is/are correct?

- 1 The dependent variable is plotted on the vertical (y) axis.
- 2 If the experiment were repeated at a higher temperature, the points plotted would lie below the original curve.
- 3 An agar block with dimensions $1\text{ cm} \times 2\text{ cm} \times 4\text{ cm}$ would take a longer time to completely decolourise than the $2\text{ cm} \times 2\text{ cm} \times 2\text{ cm}$ cube.



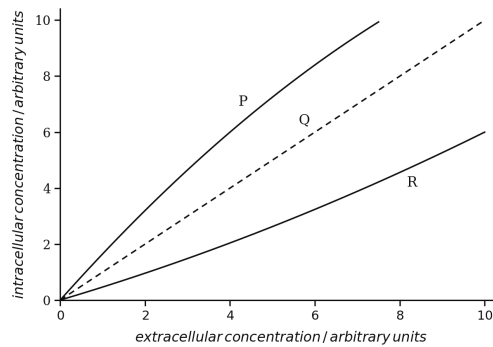
- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 9 A cell is placed in solutions containing different concentrations of three uncharged solutes: P, Q, and R. The cell membrane is permeable to all three solutes.

The graph shows the intracellular concentration of each solute once a steady state is reached, as a function of the extracellular concentration. The dashed line represents where intracellular concentration equals extracellular concentration.

Which of the following statements is/are correct?

- 1 Metabolic energy (ATP) is required to maintain the steady-state concentration of solute P.
- 2 Metabolic energy (ATP) is required to maintain the steady-state concentration of solute R.
- 3 If ATP synthesis in the cell is inhibited, the steady-state intracellular concentration of solute R will decrease.



- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 10** Two mice, both heterozygous for two independently assorting autosomal genes ($AaBb$), are crossed. Under standard Mendelian inheritance with complete dominance, the expected phenotypic ratio among their offspring is 9 : 3 : 3 : 1.

However, across a very large number of offspring from these crosses, four distinct phenotypes are consistently observed in a ratio of 6 : 3 : 2 : 1.

Which of the following statements, if true, could explain this observed phenotypic ratio?

1 Homozygous dominant individuals for gene A (AA) die before birth, while Aa and aa individuals are viable.

2 Homozygous recessive individuals for gene A (aa) die before birth, while AA and Aa individuals are viable.

3 Homozygous dominant individuals for gene B (BB) die before birth, while Bb and bb individuals are viable.

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 11** A 20-nucleotide synthetic mRNA molecule is translated in a cell-free translation system. Translation of this mRNA produces a hexapeptide (a peptide consisting of six amino acids) using one continuous reading frame from the 5' to 3' direction without any stop codons.

The sequence of the mRNA is shown below:

5'-AUUGGCAUUGACCGCAUUGC-3'

The table shows the number of each amino acid found in the resulting hexapeptide:

Amino acid	Number of residues
Leu	3
Pro	2
Met	1

What is the primary structure of the hexapeptide from the N-terminus to the C-terminus?

- A** Leu-Pro-Leu-Met-Pro-Leu
- B** Leu-Pro-Met-Leu-Pro-Leu
- C** Pro-Leu-Pro-Met-Leu-Leu
- D** Leu-Leu-Pro-Met-Pro-Leu
- E** Met-Leu-Pro-Leu-Pro-Leu
- F** Met-Pro-Leu-Pro-Leu-Leu
- G** Leu-Pro-Leu-Pro-Met-Leu
- H** Pro-Leu-Leu-Met-Leu-Pro

- 12** An experiment was conducted to investigate the gravitropic response of young primary roots. Three identical batches of seedlings were placed with their primary roots oriented horizontally (0°) at time $t = 0$ h in humid, dark chambers under the following conditions:

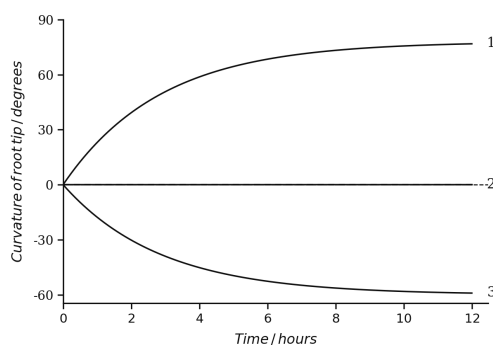
- **Treatment P:** Intact roots with no modifications (control).
- **Treatment Q:** Roots with the root cap surgically removed immediately before placement.
- **Treatment R:** Intact roots with an agar block containing a high concentration of auxin (IAA) placed only on the upper surface of the elongation zone.

The curvature of the root tips was recorded over a period of 12 hours. On the graph:

- a positive angle represents downward bending (towards gravity)
- a negative angle represents upward bending (away from gravity)
- an angle of 0° represents straight horizontal growth.

Which row of the table correctly matches each treatment to its corresponding curve on the graph?

	P	Q	R
A	1	2	3
B	1	3	2
C	2	1	3
D	2	3	1
E	3	1	2
F	3	2	1
G	1	2	1



- 13** An experiment was carried out in total darkness to investigate the gravitropic responses of young seedlings. Three identical, straight seedlings were placed horizontally at time $t = 0$ h under the following conditions:

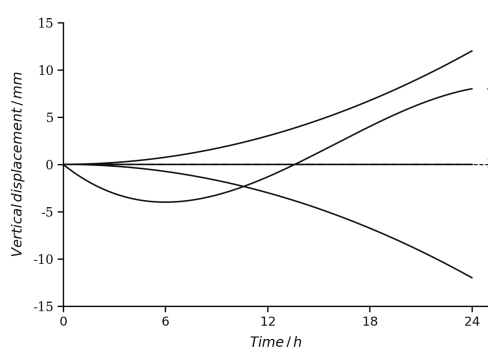
- **P**: An intact seedling kept stationary in a horizontal position; the vertical displacement of the shoot tip from the initial horizontal axis was measured over 24 hours.
- **Q**: An intact seedling kept stationary in a horizontal position; the vertical displacement of the root tip from the initial horizontal axis was measured over 24 hours.
- **R**: An intact seedling placed horizontally on a clinostat rotating slowly and continuously around its horizontal axis; the vertical displacement of the shoot tip from the initial horizontal axis was measured over 24 hours.

For all measurements, an upward displacement from the initial horizontal axis is defined as positive, and a downward displacement is defined as negative.

The graph shows four possible curves (1, 2, 3, and 4) representing vertical displacement over time.

Which row in the table correctly identifies the curves corresponding to seedlings P, Q, and R?

	P	Q	R
A	1	2	3
B	1	3	1
C	1	3	2
D	2	1	3
E	2	3	2
F	3	1	2
G	3	1	3
H	3	2	1



- 14** The diagram shows the structure of an adenosine triphosphate (ATP) molecule with components labelled 1, 2, and 3, and a specific bond labelled X.

Which of the following statements is/are correct?

- 1 Components 1 and 2, when bonded to a single phosphate group, form an adenine RNA nucleotide.
- 2 Hydrolysis of bond X produces adenosine diphosphate (ADP) and an inorganic phosphate group (P_i).
- 3 Component 2 is a pentose sugar containing one fewer oxygen atom than the pentose sugar found in DNA nucleotides.

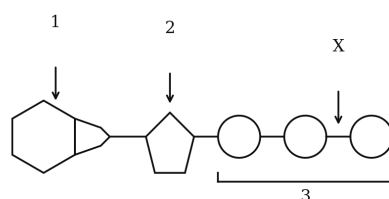


Diagram not to scale

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 15** The table below shows the mean rates of water uptake by roots and water loss by transpiration for a leafy plant over consecutive 2-hour intervals between 06:00 and 18:00 on a warm day.

Time interval	Mean rate of water uptake by roots / $\frac{\text{cm}^3}{\text{h}}$	Mean rate of water loss by transpiration / $\frac{\text{cm}^3}{\text{h}}$
06:00 – 08:00	15	10
08:00 – 10:00	25	40
10:00 – 12:00	40	65
12:00 – 14:00	50	55
14:00 – 16:00	45	30
16:00 – 18:00	25	10

Which of the following statements is/are correct?

- 1 The plant experiences its greatest internal water deficit (highest xylem tension) at 14:00.
- 2 By 18:00, the plant has fully recovered the water deficit accumulated between 08:00 and 14:00.
- 3 The water potential in the leaf mesophyll cells is lower at 14:00 than at 08:00.

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 16** The table shows characteristics of six plant species (A–F) inhabiting a fragmented landscape undergoing rapid climate change.

Which species is most at risk of extinction?

Species	Generation time	Dispersal ability	Genetic diversity	Ecological niche
A	short	high	high	generalist
B	short	low	high	specialist
C	long	low	low	specialist
D	long	high	low	generalist
E	long	low	high	generalist
F	short	low	low	specialist

- 17** A segment of double-stranded DNA is transcribed to produce an mRNA molecule containing 600 nucleotides.

The base composition of this mRNA molecule is:

- Adenine (A): 24%
- Uracil (U): 18%
- Cytosine (C): 32%
- Guanine (G): 26%

Which of the following statements about this 600-base-pair DNA segment is/are correct?

- 1 There are 192 cytosine bases in the template DNA strand.
- 2 Thymine accounts for 21% of the total nitrogenous bases in the double-stranded DNA segment.
- 3 There are 1548 hydrogen bonds between the complementary base pairs in this double-stranded DNA segment.

- A** none
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 18** In a species of moth, sex is determined by the ZW sex-chromosome system: females are heterogametic (ZW) and males are homogametic (ZZ).

A breeding experiment begins with a single mated pair of moths (Generation 0).

In each generation:

- each female mates with a male from the same generation and produces a single clutch of 10 offspring;
- the expected sex ratio of male to female offspring in each clutch is 1 : 1, and this ratio is observed in all clutches;
- all offspring survive to maturity and reproduce only within their generation.

What is the expected total number of Z chromosomes present among all individuals in the 4th generation of offspring (Generation 4)?

- A** 375
- B** 625
- C** 1250
- D** 1875
- E** 2500
- F** 3125
- G** 3750

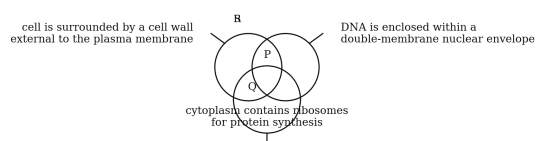
19 The Venn diagram shows three cellular features used to classify biological cells:

- Feature 1: Cell is surrounded by a cell wall external to the plasma membrane
- Feature 2: DNA is enclosed within a double-membrane nuclear envelope
- Feature 3: Cytoplasm contains ribosomes for protein synthesis

A bacterial cell, a yeast cell, and a human cheek epithelial cell can each be placed into one of the labelled regions P, Q, R, or S.

Which row in the table correctly identifies the region for each cell type?

	bacterial cell	yeast cell	human cheek epithelial cell
A	Q	S	R
B	Q	R	P
C	P	S	R
D	S	Q	P
E	S	S	R
F	R	S	Q
G	Q	P	R
H	P	Q	S



20 Root hair cells of a plant are placed in an aerated nutrient solution.

The concentrations of two dissolved ions, ion X and ion Y, and the water potential in the solution and in the root hair cell cytoplasm are:

- Ion X: concentration in solution = 0.1 mmol dm^{-3} ; concentration in cytoplasm = 4.5 mmol dm^{-3}

- Ion Y: concentration in solution = 8.0 mmol dm^{-3} ; concentration in cytoplasm = 1.2 mmol dm^{-3}

- Water potential: solution = -80 kPa ; cytoplasm = -450 kPa

Under normal aerated conditions, there is net movement of ion X, ion Y, and water into the root hair cells.

A chemical that specifically inhibits ATP synthesis is added to the nutrient solution.

Which row in the table shows the net movement of ion X, ion Y, and water across the root hair cell surface membrane immediately after the chemical is added?

	net movement of ion X	net movement of ion Y	net movement of water
A	moves into the cell	moves into the cell	moves into the cell
B	moves into the cell	moves into the cell	moves out of the cell
C	moves into the cell	does not move into the cell	moves into the cell
D	moves into the cell	does not move into the cell	moves out of the cell
E	does not move into the cell	moves into the cell	moves into the cell
F	does not move into the cell	moves into the cell	moves out of the cell
G	does not move into the cell	does not move into the cell	moves into the cell
H	does not move into the cell	does not move into the cell	moves out of the cell

- 21** The table shows several mRNA codons and the amino acids or translation functions they code for. All other codons not listed code for amino acids other than glutamic acid, aspartic acid, lysine, glutamine, or tryptophan, and none of the unlisted codons act as stop codons.

mRNA codon	Amino acid / function
GAA	glutamic acid
GAG	glutamic acid
GAC	aspartic acid
GAU	aspartic acid
AAG	lysine
CAG	glutamine
UAG	STOP
UGA	STOP
UAA	STOP
UGG	tryptophan

Consider the following six-base segment of an mRNA molecule:

5'-GAG UGG-3'

A single random base substitution mutation occurs within this six-base sequence. The mutation is equally likely to occur at any of the six nucleotide positions. At the mutated position, the original base is replaced by one of the other three RNA bases with equal probability ($\frac{1}{3}$ each).

Assuming no other mutations occur, what are the probabilities of the following outcomes?

	Probability that the mutation is silent (both codons still code for glutamic acid followed by tryptophan)	Probability that the mutation produces a stop codon in this sequence
A	$\frac{1}{18}$	$\frac{1}{18}$
B	$\frac{1}{18}$	$\frac{1}{18}$
C	$\frac{1}{18}$	$\frac{1}{18}$
D	$\frac{1}{18}$	$\frac{1}{18}$
E	$\frac{1}{18}$	$\frac{1}{18}$
F	$\frac{1}{18}$	$\frac{1}{18}$
G	$\frac{1}{18}$	$\frac{1}{18}$
H	$\frac{1}{18}$	$\frac{1}{18}$

- 22** The diagram shows a simplified cross-section through part of a dicotyledonous leaf. Consider the following statements about the two cells labelled X and Y:

- 1 Both cell X and cell Y belong to the same plant tissue.
- 2 Both cell X and cell Y contain the same nuclear genes.
- 3 Both cell X and cell Y contain chloroplasts.

Which of these statements is/are correct?

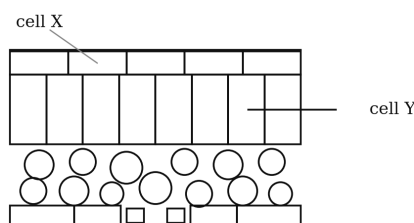


Diagram not to scale

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3
- 23** The table below displays annual carbon flux data for three different ecosystems, measured in arbitrary units of mass per area per year.
- | Ecosystem | Gross Primary Production (GPP) | Autotrophic Respiration (Ra) | Heterotrophic Respiration (Rh) |
|-----------|--------------------------------|------------------------------|--------------------------------|
| 1 | 120 | 50 | 40 |
| 2 | 100 | 40 | 65 |
| 3 | 150 | 80 | 70 |
- Which of the following conclusions regarding the carbon balance of these ecosystems is supported by the data?
- A** Ecosystem 3 acts as the largest net carbon sink because it has the highest Gross Primary Production.
- B** Ecosystem 2 acts as a net carbon source because its total respiration exceeds its Gross Primary Production.
- C** Ecosystems 1 and 3 contribute equally to the net removal of atmospheric carbon because their Net Primary Production is identical.
- D** Ecosystem 2 acts as a net carbon sink because the carbon fixed by photosynthesis exceeds the carbon released by producers.
- E** Ecosystem 1 is a net carbon source because the sum of autotrophic and heterotrophic respiration is highest relative to its GPP.

- 24** A marine biologist investigated the population of a burrowing bivalve species on a large intertidal mudflat.

A square core sampler with internal base dimensions of $0.20 \text{ m} \times 0.20 \text{ m}$ was pushed into the sediment to a uniform depth of 0.15 m . The sediment from the core was sieved to count the number of live bivalves present. This procedure was repeated 10 times.

Due to deep, soft mud further out on the mudflat, all 10 core samples were taken within 2 m of the high-tide line along the upper shore.

The number of bivalves recorded in each core is shown in the table:

Sample number	1	2	3	4	5	6	7	8	9	10
Number of bivalves	3	7	0	5	0	4	8	2	6	5

The biologist made the following statements:

- 1 Each core sampled a sediment volume of $6.0 \times 10^{-3} \text{ m}^3$.
- 2 The frequency of occurrence of the bivalves in the samples was 80%.
- 3 An accurate estimate of the total bivalve population on the entire mudflat could be calculated using these data if the total surface area of the mudflat was known.

Which of the statements about the investigation is/are correct?

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 25** In domestic cats, fur colour is controlled by an X-linked gene with two codominant alleles:
- * allele X^B produces black fur
 - * allele X^O produces orange fur
 - * heterozygous females ($X^B X^O$) have tortoiseshell fur

Two tortoiseshell female cats each mate with a different male cat:

- * Female 1 mates with a black male cat ($X^B Y$)
- * Female 2 mates with an orange male cat ($X^O Y$)

Each mating produces an equal number of male and female offspring, and both litters contain the same total number of kittens.

What is the expected proportion of all female kittens that are tortoiseshell, and what is the expected proportion of all kittens (males and females combined) across both litters that are orange?

	expected proportion of female kittens that are tortoiseshell	expected proportion of all kittens that are orange
A	0.25	0.250
B	0.25	0.375
C	0.25	0.500
D	0.50	0.250
E	0.50	0.375
F	0.50	0.500
G	0.75	0.375
H	0.75	0.500

- 26** A linear double-stranded DNA molecule contains 3000 base pairs and has a total of 7800 hydrogen bonds between its paired bases.

Which of the following statements is/are correct?

- 1 Guanine accounts for 60% of the bases in this DNA molecule.
- 2 There are 1200 adenine bases present in this DNA molecule.
- 3 A single strand of this DNA molecule contains 2999 phosphodiester bonds.

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 27 The table shows the partial pressures of oxygen (p_{ceO_2}) and carbon dioxide (p_{ceCO_2}), and the concentration of urea, in blood entering and leaving three different organs (X, Y, and Z) of a healthy human at rest:

Organ	p_{ceO_2} entering / kPa	p_{ceO_2} leaving / kPa	p_{ceCO_2} entering / kPa	p_{ceCO_2} leaving / kPa	Urea entering / mmol dm ⁻³	Urea leaving / mmol dm ⁻³
X	13.0	5.5	5.2	6.0	5.0	2.5
Y	5.5	13.0	6.0	5.2	5.0	5.0
Z	13.0	5.5	5.2	6.0	5.0	7.5

Which row in the table below correctly identifies organs X, Y, and Z?

	Organ X	Organ Y	Organ Z
A	kidney	lungs	liver
B	kidney	liver	lungs
C	lungs	kidney	liver
D	lungs	liver	kidney
E	liver	kidney	lungs
F	liver	lungs	kidney